

WASP-135b

R-1252

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_210307 · created 2026-07-31T11:56:15+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459280.9832 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.026
Transit depth (Rp/Rs)^2	2.45 +/- 0.81 [%]
Orbital Inclination (inc)	82.49 +/- 1.62
Airmass coefficient 1 (a1)	1.108 +/- 0.021
Airmass coefficient 2 (a2)	-0.074 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.95 %
Best Comparison Star	#2 - [43, 255]
Optimal Aperture	3.32
Optimal Annulus	4.85
Transit Duration (day)	0.071 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.026 vs 0.139 (deviation 0.69σ)
Duration fit vs published	0.071 d vs 0.0687 d (deviation 0.19σ)
Mid-time O–C	12.8 min
Depth SNR	6.0
Residual scatter	1.95 %

Light curve

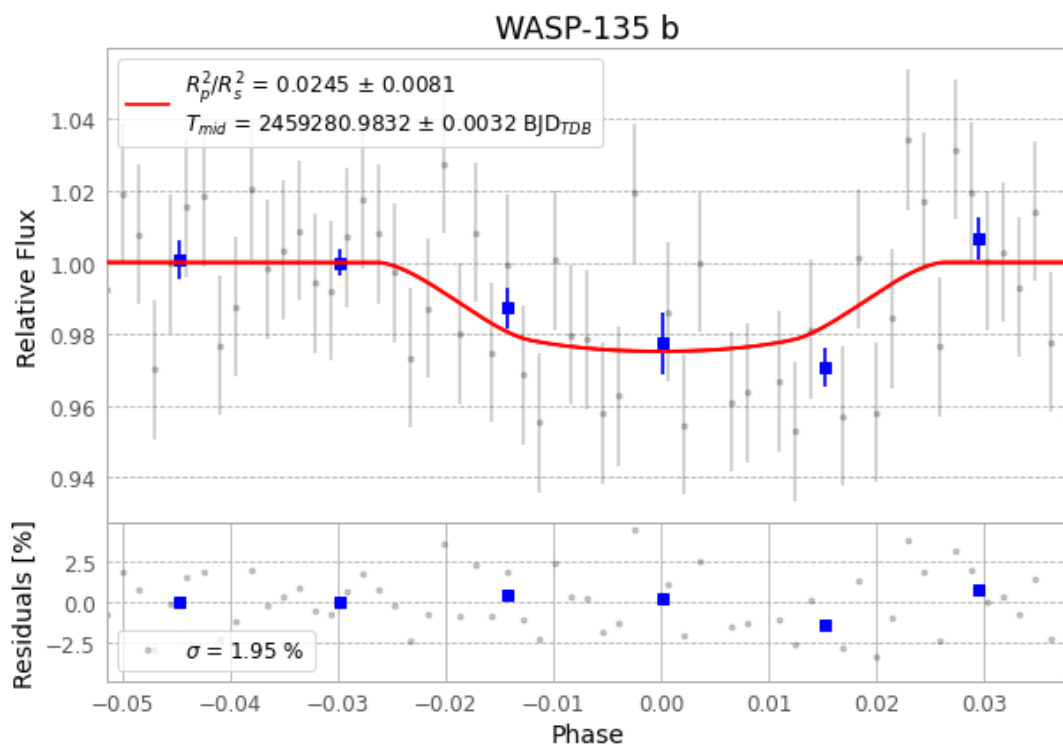


Figure 1 — FinalLightCurve_WASP-135 b_2021-03-07.png (SHA-256 97779d4d2615ceba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 379], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 370475d913c4bfe3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

61 FITS frames (40 MB), WASP-135210307095112.FITS → WASP-135210307125112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-210307.log (17a17fdd0582...), FinalLightCurve_WASP-135 b_2021-03-07.png (97779d4d2615...), FinalLightCurve_WASP-135 b_2021-03-07.csv (9dc9d753e921...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:56Z.